Jianai Zhao

AEROSPACE ENGINEERING GRADUATE, RSI-TSINGHUA'15, IBDP'17, CU BOULDER AEROENGR BS'22,
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RESEARCH INTERESTS

Fault Detection and Exclusion (FDE), Prognostics and Health Management (PHM), Data-Driven Decision Making, High Performance Computing (HPC), Artificial Intelligence/ Machine Learning (AI/ML)

PROFESSIONAL PREPARATION

MS, Aerospace Engineering	Technical University of Darmstadt	Germany	Oct 2022 – Dec 2024
BS, Aerospace Engineering Sciences	University of Colorado Boulder	USA	Aug 2017 – May 2022
International Baccalaureate Diploma	Beijing Huijia Private School	China	Sep 2014 – May 2017

PREPRINT/ CONFERENCE PAPER/ OPEN REPOSITORY

Jianai Zhao, 2024. Neural network based shock detection in compressible flow at space-re-entry conditions. Master Thesis, TU Darmstadt, Germany. [Zenodo PDF], [NASA Astrophysics Data System Indexed], [SSRN Preprint PDF]

Cameron Hahn, Roger Domingo Montaner, **Jianai Zhao**, Tobias Wetzel, and Yunfei Wang. 2023. Reverse Engineering and Machine Learning for Concept Creation of a Data Mining Hardware. Advanced Design Project Report, TU Darmstadt, Germany. [Zenodo PDF]

Shawn Stone, Walter Sabin, Tyler Gaston, Max Morgan, **Jianai Zhao**, Zack Hubbard, Jason Le, Nicholas Herrington, Jake Pirnack, and Brandon diLorenzo. 2022. CROACS: Close Range Orbital Attitude Characterization System. AIAA Region V Student Conference, Colorado Springs, USA. [Zenodo PDF] , [NASA Astrophysics Data System Indexed]

Sandor Nemethy, Sam Laurila, Rebeca Griego, Tyler Gaston, Shawn Stone, Jash Bhalavat, Nicholas Herrington, Zackary Hubbard, Jason Le, Max Morgan, Jake Pirnack, Walter Sabin, Brandon diLorenzo, **Jianai Zhao**. 2021. *ASEN 4018*. Sensors, 720: 419–8481. [Google Scholar Citation]

AWARD/ FELLOWSHIP/ MEMBERSHIP/ GRANT

Aug 2021 – May 2022	**CROACS Research Grant,** CU Boulder [Smead Aerospace] Senior Design CROACS Research Grant sponsored by [Astroscale U.S.]; managed research budget and procurement.	$5K USD
Apr 2022	**American Institute of Aeronautics and Astronautics Region V,** Student Member [PDF].	Stdnt. Mbr.
Aug 2017	**Outstanding Performance Counselor,** Research Science Initiative (RSI) at Tsinghua.	Awd.
Aug 2015	**RSI Rickoids Alumni,** Research Science Institute (RSI) [Rickoids].	Rickoid

RESEARCH EXPERIENCE

May–Dec 2024	**Master Thesis,** [Chair of Fluid Dynamics]. Thesis title: Neural Network based Shock Detection in Compressible Flow at Space-re-entry Conditions [PDF]. Advisor: [Florian Kummer]. About: Developed data-driven shock detection algorithms in Discontinuous Galerkin (DG) schemes using Gradient Boosted Decision Trees and Artificial Neural Networks. Integrated the trained AI sensors into the C#/.NET-based [BoSSS Solver] for online HPC fluid simulations. Successfully stabilized Mach 10 Double Mach Reflection flows with bounded error limits, offering a scalable machine-learning alternative to traditional threshold-based sensors such as the [Persson Sensor].	Thesis Grade: 1.0 (Highest), Germany
Oct 2023 – Mar 2024	**Boeing Global Service Jeppesen Project (Group Work),** [Institute of Flight Systems and Automatic Control]. Topic: Integration of a Steering Wheel and a Conventional Yoke in a Flight Control System. Advisor: [Jens Schiefele]. About: Cockpit design for a modular flying car with a hybrid setup using Siemens NX. The design blends driving and flying experiences, allowing both translational and rotational movement of the steering wheel to act as a flight control system.	Project Grade: 1.7, Germany

May 2023 – Sep 2023	**Mechanical Engineering Advanced Design Project (Group Work),** [Institute for Robotic Systems in Mechanical Engineering]. Title: Analysis of the Potential for Improving Velocity Predictions Using Route Information. Advisor: [Stephan Rinderknecht]. About: Researched the integration of spatial road parameters into LSTM-based vehicle velocity prediction models. Compared model accuracy using OpenStreetMap GPS queries against a custom Most Probable Path (MPP) algorithm designed for signal-deprived environments. Identified key data-conditioning constraints in map parsing and established the efficacy of path continuity in time-series forecasting.	Project Grade: 1.7, Germany
Nov 2022 – Apr 2023	**Aerospace Engineering Advanced Design Project (Group Work),** [Institute of Flight Systems and Automatic Control]. Title: Reverse Engineering and Machine Learning for Concept Creation of a Data Mining Hardware [PDF]. Advisor: [Uwe Klingauf]. About: Engineered a machine learning framework to evaluate fault tolerance in electric aircraft using high-frequency digital twin datasets. Modeled real-world sensor noise, delays, and data corruptions in MATLAB to test the robustness of multiple diagnostic classifiers. Translated model performance drop-offs into actionable hardware specifications for onboard data logging systems.	Project Grade: 1.3 (2nd-High), Germany
Oct 2022 – Mar 2023	**Machine Learning Project (Group Work),** [Institute of Flight Systems and Automatic Control]. Topic: Data-Driven System Health & Predictive Maintenance Strategy for Bulk Handling Systems. Advisor: [Uwe Klingauf]. About: Analyzed multi-year continuous bulk-handling data to identify operational degradation patterns, establishing data-synchronization protocols to support Industry 4.0 condition-based monitoring.	Project Grade: 1.3 (2nd-High), Germany
Aug 2021 – May 2022	**Bachelor Capstone Project (Group Work),** sponsored by Astroscale U.S. [Title]: Close Range Orbital Attitude Characterization System [PDF]. Advisor: [Yu Takahashi] (NASA Jet Propulsion Laboratory). Committe: [Project Advisory Board]. About: Characterized uncooperative space debris client using a scannerless LiDAR servicer and a customized 3 degree of freedom tumbling rig for ground truth. Employed 3D online template matching for initial pose acquisition and iterative closest point algorithms for dynamic tracking.	Project Grade: 4.0 (Highest), USA

OUTREACH/ PROFESSIONAL DEVELOPMENT/ SERVICE

Aug 1–4 2026	**Research Science Institute (RSI) 2026 at Massachusetts Institute of Technology,** Paper Reader. Reviewed penultimate and final papers in Aeronautics and Astronautics. [Link].	Service
Jul 27–29 2025	**Research Science Institute (RSI) 2025 at Massachusetts Institute of Technology,** Paper Reader. Reviewed penultimate and final papers in Aeronautics and Astronautics. [Link].	Service
Jun–Oct 2021	**Smith Lab at CU Boulder,** Undergrad Research Assistant (Student Assistant I). Produced science outreach videos as a part of U.S. National Science Foundation (NSF) funded ([NSF 1553114]) research and assisted in maintaining the greenhouse collections used by the lab's research. [Link].	Outreach
Jun–Jul 2018	**CU Boulder Strategic Relations and Communications,** Web Communication Intern (Student Assistant II). Creating and updating CU Boulder campus websites.	Professional Development
Jul–Aug 2017	**Research Science Initiative (RSI) 2017 at Tsinghua University,** Counselor. Supported student needs; aided social and academic adjustment; reported concerns; planned activities.	Professional Development
Jul–Aug 2015	**Research Science Initiative (RSI) 2015 at Tsinghua University,** Scholar at the School of Aerospace. Independent research in flow-induced vibration using a wind tunnel.	Professional Development